

Proportion & Rule of Three — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) $ad = bc$. Cross products are equal.

Q2 (c) $21 : 6 :: 35 : 10$. $21 \times 10 = 6 \times 35 = 210$.

Q3 30. $6x = 10 \times 18 \Rightarrow x = 30$.

Q4 (b) 15. $3 \times 20 = 4x \Rightarrow x = 15$.

Q5 True. Otherwise the cross multiplication is meaningless.

Q6 (a) Both true, R explains A. $150x = 90 \times 240 \Rightarrow x = 144$.

Short answer

Q7 Yes, proportional. $24 \times 3 = 72$ and $8 \times 9 = 72$.

Q8 30 spoons. $6 : 10 :: 18 : x \Rightarrow x = 30$.

Q9 $x = 15$. $5 \times 24 = 8x \Rightarrow x = 15$.

Long answer

Q10 144 km. $4 \text{ h} = 240 \text{ min}$; $150 : 90 :: 240 : x \Rightarrow 150x = 21600 \Rightarrow x = 144$.

Q11 10 kg. $120 : 15 :: 80 : x \Rightarrow 120x = 1200 \Rightarrow x = 10$.

Q12 Not a true proportion; the fourth term should be 9. $8 \times 6 = 48 \neq 3 \times 24 = 72$; for $8 : 3 :: 24 : ?$, $8 \times ? = 3 \times 24 \Rightarrow ? = 9$.

Total: 21 marks.